Miguel Navarro Castro

PhD Student in Mathematics · University of Seville

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RESEARCH INTERESTS

Topological data analysis (Chromatic TDA) · Mathematical and computational developmental biology · Single-cell image analysis of human pluripotent stem cells · Machine learning · Sports analytics

EDUCATION

2024 – present
PhD in Mathematics
University of Seville, Spain
Thesis: Chromatic Topological Data Analysis of BMP-directed gastrulation in hPSC colonies from fluorescence microscopy images. Advisors: María José Jiménez and Elena Camacho Aguilar.

2022 – 2024
MSc in Mathematics
University of Seville, Spain
GPA 8.57/10. Thesis: *The Bernstein–Sato ideal*. Advisor: Alberto Castaño Domínguez.

2022 – 2024
MSc in Secondary Education Teacher Training
University of Seville, Spain
GPA 8.6/10. Thesis: *Analysis and improvement proposal*. Advisor: Agustín Riscos Núñez.

2018 – 2022
BSc in Mathematics
University of Seville, Spain
GPA 6.88/10. Thesis: *The Bernstein–Sato polynomial*. Advisor: Alberto Castaño Domínguez.

GRANTS AND APPOINTMENTS

2025
IMUS Research Initiation Programme (PI3)
Institute of Mathematics (IMUS), University of Seville
Project: *Analysis of the Response of Stem Cells to Biochemical Stimuli at the Single-Cell Level*. Supervisors: María José Jiménez Rodríguez and Elena Camacho Aguilar. Cell segmentation and tracking of live-imaged, fluorescently labelled hPSCs to study how varying exposure to a differentiation-inducing protein shapes cell differentiation across experiments.

2019 – 2020
Undergraduate Research Assistant
Department of Mathematical Analysis, University of Seville

PUBLICATIONS

Peer-reviewed journal articles. * Corresponding author. JIF: Journal Impact Factor.

1. B. De-la-Cruz-Torres, A. Ruiz-de-Alarcón-Quintero, **M. Navarro-Castro**. Temporal patterns of advanced shot-quality metrics in elite men's and women's European football. *Big Data and Cognitive Computing*, 10, 214 (2026). doi.org/10.3390/bdcc10070214
2. B. De-la-Cruz-Torres, A. Ruiz-de-Alarcón-Quintero, **M. Navarro-Castro**. The role of shot velocity in advanced post-shot metrics: Evidence from the UEFA European Football Championships. *Data*, 11, 39 (2026). doi.org/10.3390/data11020039
3. B. De-la-Cruz-Torres, **M. Navarro-Castro**, A. Ruiz-de-Alarcón-Quintero. Expected Shot Impact Timing (xSIT) and other advanced metrics as indicators of performance in English men's and women's professional football. *Data*, 10(10), 159 (2025). doi.org/10.3390/data10100159
4. B. De-la-Cruz-Torres, **M. Navarro-Castro***, A. Ruiz-de-Alarcón-Quintero. Quantifying football shooting precision: The Expected Shot Impact Timing (xSIT) approach. *Applied Sciences*, 15(12), 6735 (2025). JIF 2.5, Q2. doi.org/10.3390/app15126735
5. B. De-la-Cruz-Torres, **M. Navarro-Castro***, A. Ruiz-de-Alarcón-Quintero. Analysis of shots trajectory and effectiveness in women's and men's football European Championship matches. *Big Data and Cognitive Computing*, 9(6), 157 (2025). JIF 4.4, Q1. doi.org/10.3390/bdcc9060157
6. B. De-la-Cruz-Torres, **M. Navarro-Castro**, A. Ruiz-de-Alarcón-Quintero. The influence of goalkeepers' height on soccer performance: A gender-based analysis. *International Journal of Sports Science & Coaching*, 20(2), 596–603 (2025). JIF 2.2, Q2. doi.org/10.1177/17479541241310523

7. B. De-la-Cruz-Torres, **M. Navarro-Castro***, A. Ruiz-de-Alarcón-Quintero. An Expected Goals On Target (xGOT) model: Accounting for goalkeeper performance in football. *Big Data and Cognitive Computing*, 9(3), 64 (2025). JIF 4.4, Q1. doi.org/10.3390/bdcc9030064
8. B. De-la-Cruz-Torres, **M. Navarro-Castro***, A. Ruiz-de-Alarcón-Quintero. Leveraging the Chain on Goals model in football: Applications for attack and defensive play. *Applied Sciences*, 15(2), 998 (2025). JIF 2.5, Q2. doi.org/10.3390/app15020998

CONFERENCES AND SCIENTIFIC ACTIVITIES

2026 — **Poster presentation**
Jornadas de Topología de Datos 2026

2024 — **Bringing Young Mathematicians Together, 7th edition**
17–20 November 2024

2024 — **Symposium: New Mathematical Approaches in Developmental Biology**
European Congress of Mathematics (ECM), 15–19 July 2024

TEACHING EXPERIENCE

2025 – present — **Substitute Lecturer, University of Seville**

- Mathematical Foundations for Architecture (BSc Fundamentals of Architecture), 176 h
- Structures and Foundations (MSc Architecture), 30 h
- Advanced Mathematics (BSc Health Engineering, Univ. of Málaga & Univ. of Seville), 68 h + 24 h MATLAB lab
- Numerical Modelling and Simulation (BSc Software Engineering), 12 h + 2 h MATLAB lab

2024 – 2025 — **Substitute Lecturer, University of Córdoba**

- Numerical Calculus and Statistics, 50.5 h
- Mathematical Methods I, 21 h
- Mathematical Analysis II, 63 h
- General Mathematics, 60 h

COURSES AND TRAINING

2024 — **Advanced Course on GATMAID**
Centre de Recerca Matemàtica (CRM)
Topological Data Analysis and its applications, with courses on related developments in algebra and geometry.

2024 — **VIII Bioinformatics Workshops (15 h)**
Regression techniques in R · R for RNA expression analysis · Machine learning workflows for biological applications.

Topological and Geometric Deep Learning: Theory, Methods, and Applications (7.5 h)
Doctoral course

Gromov–Hausdorff Distance and Stability Results in Computational Topology (6 h)
Workshop

Machine Learning and Big Data for Bioinformatics, 4th ed. (100 h)
MOOC, University of Granada

Google Cloud: Associate Cloud Engineer (40 h) and Data Analyst (30 h)
Continuing Education, University of Seville

2022 — **Python for Beginners (30 h)**
Online course

SKILLS

Programming — Python (scikit-learn and the scientific Python stack), MATLAB, R, SQL, HTML, LaTeX, Maxima, GeoGebra

Languages — Spanish (native) · English (B2, Trinity College London)